

ST459 Summative: Quantum Computing and Information

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Conventions

I use Dirac notation for computational basis states: $|0\rangle, |1\rangle$ (and similarly $|a\rangle$ for n -bit basis states). Inner products are linear in the second argument and conjugate-linear in the first.

Exercise 1 (Equivalent characterisations of unitarity)

Let $U : \mathcal{B} \rightarrow \mathcal{B}$ be linear and unitary, i.e. $\|U\beta\| = \|\beta\|$ for all $\beta \in \mathcal{B}$.

1.1 Show that U is invertible, and U^{-1} is unitary

Invertibility. If U were not injective, there would exist $\alpha \neq 0$ with $U\alpha = 0$. Then

$$\|\alpha\| = \|U\alpha\| = \|0\| = 0,$$

a contradiction. Since $\mathcal{B}$ is finite-dimensional, injective $\iff$ surjective, so U is invertible.

U^{-1} is unitary. Let $\beta \in \mathcal{B}$, and put $\alpha = U^{-1}\beta$. Then $\beta = U\alpha$, so

$$\|U^{-1}\beta\| = \|\alpha\| = \|U\alpha\| = \|\beta\|.$$

Therefore U^{-1} preserves norms, i.e. U^{-1} is unitary.

1.2 Show that U preserves all inner products

We are given $\|U\gamma\|^2 = \|\gamma\|^2$ for all γ .

The complex polarisation identity is

$$\langle \alpha, \beta \rangle = \frac{1}{4} \sum_{k=0}^3 i^k \left\| \alpha + i^k \beta \right\|^2.$$

It is sufficient to recover the real and imaginary parts of $\langle U\alpha, U\beta \rangle$ from norm preservation; the following computations do exactly that.

Take arbitrary $\alpha, \beta \in \mathcal{B}$.

Recovering $\operatorname{Re}\langle \cdot, \cdot \rangle$. From $\gamma = \alpha + \beta$:

$$\|U(\alpha + \beta)\|^2 = \|\alpha + \beta\|^2.$$

Expanding both sides,

$$\|U\alpha\|^2 + 2\operatorname{Re}\langle U\alpha, U\beta \rangle + \|U\beta\|^2 = \|\alpha\|^2 + 2\operatorname{Re}\langle \alpha, \beta \rangle + \|\beta\|^2.$$

Using norm preservation $\|U\alpha\| = \|\alpha\|$ and $\|U\beta\| = \|\beta\|$, I obtain

$$\operatorname{Re}\langle U\alpha, U\beta \rangle = \operatorname{Re}\langle \alpha, \beta \rangle.$$

Recovering $\text{Im}\langle \cdot, \cdot \rangle$. Now take $\gamma = \alpha + i\beta$:

$$\|U(\alpha + i\beta)\|^2 = \|\alpha + i\beta\|^2.$$

Expanding,

$$\|U\alpha\|^2 - 2\text{Im}\langle U\alpha, U\beta \rangle + \|U\beta\|^2 = \|\alpha\|^2 - 2\text{Im}\langle \alpha, \beta \rangle + \|\beta\|^2,$$

so

$$\text{Im}\langle U\alpha, U\beta \rangle = \text{Im}\langle \alpha, \beta \rangle.$$

Equality of real and imaginary parts implies

$$\boxed{\langle U\alpha, U\beta \rangle = \langle \alpha, \beta \rangle \quad \forall \alpha, \beta \in \mathcal{B}.}$$

1.3 Show that orthonormal image of basis states is also sufficient for unitarity

Assume now that $U : \mathcal{B} \rightarrow \mathcal{B}$ is linear and satisfies

$$\langle U(|a\rangle), U(|b\rangle) \rangle = \mathbf{1}_{\{a=b\}} \quad \forall a, b \in \{0, 1\}^n.$$

I show U is unitary.

Let

$$\alpha = \sum_a \alpha_a |a\rangle, \quad \beta = \sum_b \beta_b |b\rangle.$$

By linearity and conjugate-linearity of the inner product,

$$\begin{aligned} \langle U\alpha, U\beta \rangle &= \left\langle \sum_a \alpha_a U(|a\rangle), \sum_b \beta_b U(|b\rangle) \right\rangle \\ &= \sum_{a,b} \overline{\alpha_a} \beta_b \langle U(|a\rangle), U(|b\rangle) \rangle \\ &= \sum_{a,b} \overline{\alpha_a} \beta_b \mathbf{1}_{\{a=b\}} \\ &= \sum_a \overline{\alpha_a} \beta_a = \langle \alpha, \beta \rangle. \end{aligned}$$

Therefore U preserves all inner products. In particular,

$$\|U\alpha\|^2 = \langle U\alpha, U\alpha \rangle = \langle \alpha, \alpha \rangle = \|\alpha\|^2 \quad \forall \alpha \in \mathcal{B}.$$

Therefore U preserves norms, so U is unitary.

A linear operator is unitary iff it maps the computational basis to an orthonormal basis.

Exercise 2 (Pauli operators)

Recall the Pauli operators on the single-qubit space:

$$\sigma_0 = I, \quad \sigma_1 = X, \quad \sigma_2 = Y, \quad \sigma_3 = Z,$$

with actions

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle, \quad Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle.$$

2.1 Show that $\sigma_j^2 = \text{id}$ for all $j = 0, 1, 2, 3$

It is sufficient to check on basis states $|0\rangle, |1\rangle$ and to use linearity.

- $\sigma_0^2 = I^2 = I$ trivially.
- $X^2 |0\rangle = X |1\rangle = |0\rangle$ and $X^2 |1\rangle = X |0\rangle = |1\rangle$, so $X^2 = I$.
- $Z^2 |0\rangle = Z |0\rangle = |0\rangle$ and $Z^2 |1\rangle = Z(-|1\rangle) = |1\rangle$, so $Z^2 = I$.
- $Y^2 |0\rangle = Y(i|1\rangle) = iY|1\rangle = i(-i|0\rangle) = |0\rangle$ and

$$Y^2 |1\rangle = Y(-i|0\rangle) = -iY|0\rangle = -i(i|1\rangle) = |1\rangle,$$

so $Y^2 = I$.

Therefore

$$\sigma_j^2 = \text{id for all } j = 0, 1, 2, 3.$$

2.2 Show that all σ_j are unitary

I use Exercise 1.3: it is sufficient to show that each σ_j maps the computational basis to an orthonormal basis.

From the defining actions,

$$I|0\rangle = |0\rangle, I|1\rangle = |1\rangle, \quad X|0\rangle = |1\rangle, X|1\rangle = |0\rangle,$$

$$Y|0\rangle = i|1\rangle, Y|1\rangle = -i|0\rangle, \quad Z|0\rangle = |0\rangle, Z|1\rangle = -|1\rangle.$$

In each case, the images of $|0\rangle, |1\rangle$ are orthonormal: they are either $|0\rangle, |1\rangle$ in some order, multiplied by phases of modulus 1.

Therefore each σ_j maps the computational basis to an orthonormal basis, so by Exercise 1.3 each σ_j is unitary:

$$\sigma_j \text{ is unitary for } j = 0, 1, 2, 3.$$

2.3 If $U^2 = \text{id}$, show eigenvalues lie in $\{-1, 1\}$

Let $U\beta = \lambda\beta$ for some nonzero eigenvector β (normalised if desired). Apply U again:

$$\beta = U^2\beta = U(\lambda\beta) = \lambda U\beta = \lambda^2\beta.$$

Since $\beta \neq 0$, I get $\lambda^2 = 1$, therefore

$$\lambda \in \{-1, 1\}.$$

This applies in particular to all Pauli operators.

2.4 Eigenvectors of σ_j for $j = 0, 1, 2, 3$

I solve the eigenvalue equation in each case. These eigenvectors are exactly the standard X -, Y -, and Z -basis states used throughout quantum computing.

$\sigma_0 = I$. Every unit vector is an eigenvector with eigenvalue $+1$. (There is no eigenvalue -1 .)

$\sigma_1 = X$. Solve

$$X(\alpha|0\rangle + \beta|1\rangle) = \beta|0\rangle + \alpha|1\rangle = \lambda(\alpha|0\rangle + \beta|1\rangle).$$

For $\lambda = +1$, we need $\alpha = \beta$, giving eigenvector

$$|\phi_+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle).$$

For $\lambda = -1$, we need $\beta = -\alpha$, giving eigenvector

$$|\phi_-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

$\sigma_2 = Y$. Let $v = \alpha|0\rangle + \gamma|1\rangle$. Using

$$Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle,$$

I get

$$Y(\alpha|0\rangle + \gamma|1\rangle) = (-i\gamma)|0\rangle + (i\alpha)|1\rangle.$$

For eigenvalue $+1$:

$$-i\gamma = \alpha, \quad i\alpha = \gamma \implies \gamma = i\alpha,$$

so an eigenvector is

$$|\phi_{+i}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle).$$

For eigenvalue -1 :

$$-i\gamma = -\alpha, \quad i\alpha = -\gamma \implies \gamma = -i\alpha,$$

so an eigenvector is

$$|\phi_{-i}\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle).$$

$\sigma_3 = Z$. I have

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

Therefore $|0\rangle$ is an eigenvector with eigenvalue $+1$, and $|1\rangle$ is an eigenvector with eigenvalue -1 .

So the eigenvectors/eigenvalues are:

σ_0 : all vectors eigenvalue $+1$, X : $\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$ $(+1)$, $\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$ (-1) , Y : $\frac{ 0\rangle + i 1\rangle}{\sqrt{2}}$ $(+1)$, $\frac{ 0\rangle - i 1\rangle}{\sqrt{2}}$ (-1) , Z : $ 0\rangle$ $(+1)$, $ 1\rangle$ (-1) .

In each case (X, Y, Z) , the listed eigenvectors form an orthonormal basis of $\mathbb{C}^2$, so the operator admits a spectral decomposition and is diagonalisable.

Exercise 3 (Single-qubit parametrisation and involution)

Let β be a qubit state (unit vector) not equal to $|0\rangle$ up to global phase.

3.1 Parametrisation up to global phase

I show that such a state can be written (up to global phase) as

$$\beta \sim \sqrt{1-q} |0\rangle + e^{i\theta} \sqrt{q} |1\rangle, \quad 0 < q \leq 1, \quad \theta \in \mathbb{R}.$$

($\Rightarrow$) Let $\beta = \beta_0 |0\rangle + \beta_1 |1\rangle$ with $|\beta_0|^2 + |\beta_1|^2 = 1$. Since β is *not* equal to $|0\rangle$ up to global phase, I must have $\beta_1 \neq 0$.

Set

$$q := |\beta_1|^2 \in (0, 1], \quad |\beta_0| = \sqrt{1-q}.$$

Choose a phase $\xi \in \mathbb{R}$ such that $e^{-i\xi} \beta_0 = \sqrt{1-q}$ (this is possible, and if $\beta_0 = 0$ then both sides are 0). Then $|e^{-i\xi} \beta_1| = \sqrt{q}$, so there exists $\theta \in \mathbb{R}$ such that

$$e^{-i\xi} \beta_1 = e^{i\theta} \sqrt{q}.$$

Therefore

$$e^{-i\xi} \beta = \sqrt{1-q} |0\rangle + e^{i\theta} \sqrt{q} |1\rangle.$$

($\Leftarrow$) Any vector of the displayed form has norm

$$(1-q) + q = 1,$$

and because $q > 0$ its $|1\rangle$ -coefficient is nonzero, so it cannot be a global phase multiple of $|0\rangle$.

Therefore the characterisation holds.

3.2 Construct the unique linear operator U with $U(|0\rangle) = \beta$ and $U(\beta) = |0\rangle$

Write

$$\beta = \sqrt{1-q} |0\rangle + e^{i\theta} \sqrt{q} |1\rangle, \quad 0 < q \leq 1.$$

Suppose U is linear and satisfies

$$U|0\rangle = \beta, \quad U\beta = |0\rangle.$$

Since $\beta = \sqrt{1-q} |0\rangle + e^{i\theta} \sqrt{q} |1\rangle$, linearity gives

$$U\beta = \sqrt{1-q} U|0\rangle + e^{i\theta} \sqrt{q} U|1\rangle = \sqrt{1-q} \beta + e^{i\theta} \sqrt{q} U|1\rangle.$$

Set this equal to $|0\rangle$:

$$|0\rangle = \sqrt{1-q} \beta + e^{i\theta} \sqrt{q} U|1\rangle.$$

Substitute β :

$$\sqrt{1-q} \beta = (1-q) |0\rangle + e^{i\theta} \sqrt{q(1-q)} |1\rangle.$$

Therefore

$$e^{i\theta} \sqrt{q} U|1\rangle = q |0\rangle - e^{i\theta} \sqrt{q(1-q)} |1\rangle.$$

Divide by $e^{i\theta} \sqrt{q}$:

$$\boxed{U|1\rangle = e^{-i\theta} \sqrt{q} |0\rangle - \sqrt{1-q} |1\rangle.}$$

Uniqueness. The condition $U|0\rangle = \beta$ fixes the image of $|0\rangle$. The additional condition $U\beta = |0\rangle$ and linearity force a single value of $U|1\rangle$, as shown in the computation above. A linear operator on a 2D space is uniquely determined by its action on a basis, so the resulting U is unique.

3.3 Show that U is unitary and $U^2 = \text{id}$

It is sufficient to show $\{\beta, \delta\}$ is orthonormal, so by Exercise 1.3 the map U is unitary; then I verify $U^2 = \text{id}$ by checking it on $|0\rangle$ and $|1\rangle$.

From part 3.2,

$$U|0\rangle = \beta, \quad U|1\rangle = \delta := e^{-i\theta}\sqrt{q}|0\rangle - \sqrt{1-q}|1\rangle.$$

I first show that $\{\beta, \delta\}$ is orthonormal. Since

$$\beta = \sqrt{1-q}|0\rangle + e^{i\theta}\sqrt{q}|1\rangle,$$

I have

$$\|\beta\|^2 = (1-q) + q = 1, \quad \|\delta\|^2 = q + (1-q) = 1.$$

Also,

$$\langle\beta, \delta\rangle = \langle\sqrt{1-q}|0\rangle + e^{i\theta}\sqrt{q}|1\rangle, e^{-i\theta}\sqrt{q}|0\rangle - \sqrt{1-q}|1\rangle\rangle.$$

Since the inner product is conjugate-linear in the first slot,

$$\begin{aligned} \langle\beta, \delta\rangle &= \overline{\sqrt{1-q}}(e^{-i\theta}\sqrt{q})\langle 0|, |0\rangle\rangle + \overline{\sqrt{1-q}}(-\sqrt{1-q})\langle 0|, |1\rangle\rangle \\ &\quad + \overline{e^{i\theta}\sqrt{q}}(e^{-i\theta}\sqrt{q})\langle 1|, |0\rangle\rangle + \overline{e^{i\theta}\sqrt{q}}(-\sqrt{1-q})\langle 1|, |1\rangle\rangle \\ &= \sqrt{1-q}e^{-i\theta}\sqrt{q} + 0 + 0 + e^{-i\theta}\sqrt{q}(-\sqrt{1-q}) \\ &= e^{-i\theta}\sqrt{q(1-q)} - e^{-i\theta}\sqrt{q(1-q)} \\ &= 0. \end{aligned}$$

Therefore $\{\beta, \delta\}$ is an orthonormal basis.

Since U maps the orthonormal basis $\{|0\rangle, |1\rangle\}$ to the orthonormal basis $\{\beta, \delta\}$, Exercise 1.3 implies that U is unitary.

There is left to show that $U^2 = \text{id}$. We already know

$$U^2|0\rangle = U\beta = |0\rangle.$$

A direct inversion of the basis change gives the decomposition of $|0\rangle$ in the $\{\beta, \delta\}$ -basis:

$$\begin{aligned} \sqrt{1-q}\beta + e^{i\theta}\sqrt{q}\delta &= \sqrt{1-q}(\sqrt{1-q}|0\rangle + e^{i\theta}\sqrt{q}|1\rangle) + e^{i\theta}\sqrt{q}(e^{-i\theta}\sqrt{q}|0\rangle - \sqrt{1-q}|1\rangle) \\ &= ((1-q) + q)|0\rangle + (e^{i\theta}\sqrt{q(1-q)} - e^{i\theta}\sqrt{q(1-q)})|1\rangle \\ &= |0\rangle. \end{aligned}$$

So

$$|0\rangle = \sqrt{1-q}\beta + e^{i\theta}\sqrt{q}\delta.$$

Rearranging,

$$\delta = \frac{1}{e^{i\theta}\sqrt{q}}(|0\rangle - \sqrt{1-q}\beta).$$

Applying U gives

$$U\delta = \frac{1}{e^{i\theta}\sqrt{q}}(U|0\rangle - \sqrt{1-q}U\beta) = \frac{1}{e^{i\theta}\sqrt{q}}(\beta - \sqrt{1-q}|0\rangle).$$

Using the expression for β ,

$$\beta - \sqrt{1-q}|0\rangle = e^{i\theta}\sqrt{q}|1\rangle,$$

so

$$U\delta = |1\rangle.$$

Therefore

$$U^2 |1\rangle = U(U|1\rangle) = U\delta = |1\rangle.$$

Since U^2 agrees with id on the basis $\{|0\rangle, |1\rangle\}$, I conclude that

$$\boxed{U^2 = \text{id.}}$$

Exercise 4 (Amplitude amplification/Grover generalisation)

We are given

$$\psi := A(|0\rangle^{\otimes n}) = \sqrt{1-p}\beta + \sqrt{p}\gamma, \quad p \in (0, 1),$$

and define

$$\theta := \arcsin(\sqrt{p}) \in \left(0, \frac{\pi}{2}\right).$$

Therefore

$$\sqrt{1-p} = \cos \theta, \quad \sqrt{p} = \sin \theta,$$

so

$$\psi = \cos \theta \beta + \sin \theta \gamma.$$

Also Z_f flips the phase of good states and fixes bad states, so on $\text{span}\{\beta, \gamma\}$:

$$Z_f \beta = \beta, \quad Z_f \gamma = -\gamma.$$

Define

$$G := AZ_{\text{OR}}A^{-1}Z_f.$$

4.1 Show that $AZ_{\text{OR}}A^{-1} = 2P_\psi - \text{id}$

The key idea is to conjugate the reflection about $|0\rangle^{\otimes n}$ by A , which turns it into the reflection about $\psi = A|0\rangle^{\otimes n}$.

Recall

$$Z_{\text{OR}} = 2P_{|0\rangle^{\otimes n}} - \text{id}.$$

Therefore

$$AZ_{\text{OR}}A^{-1} = A(2P_{|0\rangle^{\otimes n}} - \text{id})A^{-1} = 2(AP_{|0\rangle^{\otimes n}}A^{-1}) - \text{id}.$$

So it is sufficient to show

$$AP_{|0\rangle^{\otimes n}}A^{-1} = P_\psi.$$

Let $v = \psi$. Then $A^{-1}\psi = |0\rangle^{\otimes n}$, so

$$AP_{|0\rangle^{\otimes n}}A^{-1}\psi = AP_{|0\rangle^{\otimes n}}(|0\rangle^{\otimes n}) = A(|0\rangle^{\otimes n}) = \psi = P_\psi\psi.$$

Now let $v \perp \psi$. Since A is unitary, $A^{-1}v \perp A^{-1}\psi = |0\rangle^{\otimes n}$, so

$$P_{|0\rangle^{\otimes n}}A^{-1}v = 0.$$

Therefore

$$AP_{|0\rangle^{\otimes n}}A^{-1}v = 0 = P_\psi v.$$

The two operators agree on ψ and on every vector orthogonal to ψ , and so on the whole space by linearity. Therefore

$$\boxed{AZ_{\text{OR}}A^{-1} = 2P_\psi - \text{id.}}$$

4.2 Show the action of G on β, γ

The crucial observation is that G acts as a rotation on the two-dimensional space $\text{span}\{\beta, \gamma\}$.

Both factors in $G = (2P_\psi - \text{id})Z_f$ preserve $\text{span}\{\beta, \gamma\}$: Z_f sends $\beta \mapsto \beta$ and $\gamma \mapsto -\gamma$, and $2P_\psi - \text{id}$ is the reflection about the line spanned by $\psi = \cos \theta \beta + \sin \theta \gamma$, which lies in $\text{span}\{\beta, \gamma\}$. Therefore G preserves $\text{span}\{\beta, \gamma\}$, so restricting attention to that plane is justified.

Using part 4.1,

$$G = (2P_\psi - \text{id})Z_f.$$

On $\text{span}\{\beta, \gamma\}$ I have $\psi = \cos \theta \beta + \sin \theta \gamma$, so

$$P_\psi v = \langle \psi, v \rangle \psi.$$

Since β, γ are orthonormal,

$$P_\psi \beta = \cos \theta \psi, \quad P_\psi \gamma = \sin \theta \psi.$$

Also $Z_f \beta = \beta$, so

$$\begin{aligned} G\beta &= (2P_\psi - \text{id})\beta = 2\cos \theta \psi - \beta \\ &= 2\cos \theta (\cos \theta \beta + \sin \theta \gamma) - \beta \\ &= (2\cos^2 \theta - 1)\beta + 2\sin \theta \cos \theta \gamma \\ &= \cos(2\theta)\beta + \sin(2\theta)\gamma. \end{aligned}$$

And $Z_f \gamma = -\gamma$, so

$$\begin{aligned} G\gamma &= (2P_\psi - \text{id})(-\gamma) = -2\sin \theta \psi + \gamma \\ &= -2\sin \theta (\cos \theta \beta + \sin \theta \gamma) + \gamma \\ &= -2\sin \theta \cos \theta \beta + (1 - 2\sin^2 \theta)\gamma \\ &= -\sin(2\theta)\beta + \cos(2\theta)\gamma. \end{aligned}$$

Therefore

$G\beta = \cos(2\theta)\beta + \sin(2\theta)\gamma, \quad G\gamma = -\sin(2\theta)\beta + \cos(2\theta)\gamma.$

4.3 Show $\psi^{(k)} = G^k A |0\rangle^{\otimes n} = \cos((2k+1)\theta)\beta + \sin((2k+1)\theta)\gamma$

I prove by induction on $k \in \mathbb{N}$.

Base case $k = 0$:

$$\psi^{(0)} = A |0\rangle^{\otimes n} = \psi = \cos \theta \beta + \sin \theta \gamma,$$

which is the formula with $2k+1 = 1$.

Inductive step: Assume

$$\psi^{(k)} = \cos((2k+1)\theta)\beta + \sin((2k+1)\theta)\gamma.$$

Then, using part 4.2,

$$\begin{aligned}
\psi^{(k+1)} &= G\psi^{(k)} \\
&= \cos((2k+1)\theta)G\beta + \sin((2k+1)\theta)G\gamma \\
&= \cos((2k+1)\theta)(\cos(2\theta)\beta + \sin(2\theta)\gamma) + \sin((2k+1)\theta)(-\sin(2\theta)\beta + \cos(2\theta)\gamma) \\
&= (\cos((2k+1)\theta)\cos(2\theta) - \sin((2k+1)\theta)\sin(2\theta))\beta \\
&\quad + (\sin((2k+1)\theta)\cos(2\theta) + \cos((2k+1)\theta)\sin(2\theta))\gamma \\
&= \cos((2k+1)\theta + 2\theta)\beta + \sin((2k+1)\theta + 2\theta)\gamma \\
&= \cos((2k+3)\theta)\beta + \sin((2k+3)\theta)\gamma.
\end{aligned}$$

Therefore, for all k ,

$$\boxed{\psi^{(k)} = \cos((2k+1)\theta)\beta + \sin((2k+1)\theta)\gamma.}$$

4.4 Add an ancilla and apply $U_{\tilde{f}}$

Observe that that $U_{\tilde{f}}$ writes the value of f into the ancilla, so bad and good components become correlated with $|0\rangle$ and $|1\rangle$ respectively.

We are given

$$U_{\tilde{f}}(|a\rangle \otimes |b\rangle) = |a\rangle \otimes |b \oplus f(a)\rangle.$$

By definition of β (uniform bad superposition) and γ (uniform good superposition),

$$U_{\tilde{f}}(\beta \otimes |0\rangle) = \beta \otimes |0\rangle, \quad U_{\tilde{f}}(\gamma \otimes |0\rangle) = \gamma \otimes |1\rangle.$$

Using part 4.3 and linearity,

$$\begin{aligned}
U_{\tilde{f}}(\psi^{(k)} \otimes |0\rangle) &= U_{\tilde{f}}\left(\cos((2k+1)\theta)(\beta \otimes |0\rangle) + \sin((2k+1)\theta)(\gamma \otimes |0\rangle)\right) \\
&= \cos((2k+1)\theta)(\beta \otimes |0\rangle) + \sin((2k+1)\theta)(\gamma \otimes |1\rangle).
\end{aligned}$$

So

$$\boxed{U_{\tilde{f}}(\psi^{(k)} \otimes |0\rangle) = \cos((2k+1)\theta)(\beta \otimes |0\rangle) + \sin((2k+1)\theta)(\gamma \otimes |1\rangle).}$$

4.5 Measuring only the ancilla

After part 4.4 the state is

$$\cos((2k+1)\theta)(\beta \otimes |0\rangle) + \sin((2k+1)\theta)(\gamma \otimes |1\rangle).$$

Now measure only the ancilla in the computational basis $\{|0\rangle, |1\rangle\}$; this is a projective measurement in that basis, with projectors

$$\text{id} \otimes P_{|0\rangle}, \quad \text{id} \otimes P_{|1\rangle}.$$

The ancilla outcome is:

- 0 with probability $\cos^2((2k+1)\theta)$, and then the first n qubits collapse to β ;
- 1 with probability $\sin^2((2k+1)\theta)$, and then the first n qubits collapse to γ .

Each post measurement state is normalised: $\|\beta\| = \|\gamma\| = 1$, so the collapsed first-register state is exactly β or γ (rather than a renormalised scalar multiple). Because the ancilla was measured, it is *known* which outcome occurred. Therefore it is known whether the first register is in β (if ancilla = 0) or in γ (if ancilla = 1).

4.6 Choosing k to maximise success probability

The success probability is

$$P_{\text{succ}}(k) = \sin^2((2k+1)\theta).$$

This is maximised when $(2k+1)\theta$ is as close as possible to an odd multiple of $\pi/2$; the first maximum occurs near

$$(2k+1)\theta \approx \frac{\pi}{2}.$$

So I choose

$$k \approx \frac{\pi}{4\theta} - \frac{1}{2}.$$

A standard integer choice is

$$\boxed{k = \left\lfloor \frac{\pi}{4\theta} \right\rfloor} \quad (\text{or the nearest integer to } \frac{\pi}{4\theta} - \frac{1}{2}).$$

With such a choice, $(2k+1)\theta$ is close to $\pi/2$, so $\sin^2((2k+1)\theta)$ is close to 1 (not only maximised).

There is left to justify the asymptotic size of k in terms of p . Since

$$\theta = \arcsin(\sqrt{p}), \quad 0 < \sqrt{p} < 1,$$

I use the bounds

$$x \leq \arcsin x \leq \frac{\pi}{2}x, \quad x \in [0, 1].$$

For completeness: the lower bound follows because $\sin y \leq y$ for $y \geq 0$, so with $y = \arcsin x$ I get

$$x = \sin(\arcsin x) \leq \arcsin x.$$

For the upper bound, $\arcsin$ is concave on $[0, 1]$, with

$$\arcsin(0) = 0, \quad \arcsin(1) = \frac{\pi}{2}.$$

Therefore its graph lies below the secant line joining $(0, 0)$ and $(1, \pi/2)$, namely $y = (\pi/2)x$, so

$$\arcsin x \leq \frac{\pi}{2}x \quad (0 \leq x \leq 1).$$

Applying this with $x = \sqrt{p}$ gives

$$\sqrt{p} \leq \theta \leq \frac{\pi}{2}\sqrt{p}.$$

Therefore $\theta = \Theta(\sqrt{p})$, and therefore

$$\frac{1}{\theta} = \Theta\left(\frac{1}{\sqrt{p}}\right).$$

More explicitly, for the choice $k = \left\lfloor \frac{\pi}{4\theta} \right\rfloor$:

$$k \leq \frac{\pi}{4\theta} \leq \frac{\pi}{4\sqrt{p}},$$

using $\theta \geq \sqrt{p}$, and

$$k \geq \frac{\pi}{4\theta} - 1 \geq \frac{\pi}{4 \cdot (\pi/2)\sqrt{p}} - 1 = \frac{1}{2\sqrt{p}} - 1,$$

using $\theta \leq \frac{\pi}{2}\sqrt{p}$. These bounds show k is bounded above and below by constant multiples of $p^{-1/2}$ (up to an additive constant), so

$$\boxed{k = \Theta(p^{-1/2}).}$$

Exercise 5 (Controlled powers U^k)

Let the first register have t qubits and the second register have n qubits. Let $T = 2^t$, and identify each t -bit string

$$k_1 k_2 \cdots k_t \in \{0, 1\}^t$$

with the integer

$$k = \sum_{r=1}^t k_r 2^{t-r} \in \mathbb{Z}_T = \{0, \dots, T-1\}.$$

Define the operator C by

$$C(|k\rangle \otimes |b\rangle) = |k\rangle \otimes U^k |b\rangle, \quad (k, b) \in \mathbb{Z}_T \times \mathbb{Z}_N,$$

and extend linearly.

5.1 Prove directly that C is unitary

By Exercise 1.3, it is enough to show that the images of computational basis states form an orthonormal family.

Take basis product states $|k\rangle \otimes |b\rangle$ and $|\ell\rangle \otimes |c\rangle$. Then

$$C(|k\rangle \otimes |b\rangle) = |k\rangle \otimes U^k |b\rangle, \quad C(|\ell\rangle \otimes |c\rangle) = |\ell\rangle \otimes U^\ell |c\rangle.$$

Using the tensor inner product identity,

$$\begin{aligned} \langle C(|k\rangle \otimes |b\rangle), C(|\ell\rangle \otimes |c\rangle) \rangle &= \langle |k\rangle \otimes U^k |b\rangle, |\ell\rangle \otimes U^\ell |c\rangle \rangle \\ &= \langle |k\rangle, |\ell\rangle \rangle \langle U^k |b\rangle, U^\ell |c\rangle \rangle. \end{aligned}$$

If $k \neq \ell$, this is 0 because $\langle |k\rangle, |\ell\rangle \rangle = 0$.

If $k = \ell$, then U^k is unitary (as a composition of unitary operators), so by Exercise 1.2 it preserves inner products:

$$\langle U^k |b\rangle, U^k |c\rangle \rangle = \langle |b\rangle, |c\rangle \rangle.$$

Therefore

$$\langle C(|k\rangle \otimes |b\rangle), C(|\ell\rangle \otimes |c\rangle) \rangle = \mathbf{1}_{\{k=\ell\}} \mathbf{1}_{\{b=c\}}.$$

So the images of basis states are orthonormal. Therefore C is unitary.

$C \text{ is unitary.}$

5.2 Show the circuit with controlled U^{2^j} realises C

Let the control register bits be $k_1 k_2 \cdots k_t$, identified with

$$k = \sum_{r=1}^t k_r 2^{t-r}.$$

The circuit applies, for each control qubit r , a controlled- $U^{2^{t-r}}$ gate.

On a basis state $|k\rangle \otimes |b\rangle$:

- if $k_r = 0$, the controlled- $U^{2^{t-r}}$ does nothing;

- if $k_r = 1$, it applies $U^{2^{t-r}}$ to the second register.

Therefore the total action on the second register is

$$\prod_{r=1}^t \left(U^{2^{t-r}} \right)^{k_r}.$$

These controlled gates commute with each other: each gate is block-diagonal in the computational basis of the control register, and on any fixed basis state $|k\rangle$ the target action is multiplication by a power of U , so the order of application does not change the resulting product on the target register. Here the factors commute, since for any a, b one has $U^a U^b = U^{a+b} = U^b U^a$. Therefore

$$\prod_{r=1}^t \left(U^{2^{t-r}} \right)^{k_r} = U^{\sum_{r=1}^t k_r 2^{t-r}} = U^k.$$

Therefore the circuit maps

$$|k\rangle \otimes |b\rangle \longmapsto |k\rangle \otimes U^k |b\rangle,$$

which is exactly the operator C .

The controlled- $U^{2^{t-r}}$ cascade implements $|k\rangle \otimes |b\rangle \mapsto |k\rangle \otimes U^k |b\rangle$.